Yihuan Huang

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EDUCATION

Robotics Engineering, Chongqing University, Chongqing, China Sep 2023 – Jun 2027

- WES GPA: 3.75 / 4.0; Overall GPA: 3.65 / 4.0 (87.62 / 100)
- Core Courses: Software Design (92/100), Quantitative Engineering Analysis 1 (96/100), Engineering Principles (90/100), Engineering Design (90/100), Integrated Mathematics and Physics (96/100)

Winter Short Programme in Artificial Intelligence and Entrepreneurship Innovation, Technical University of Munich, Germany Jan 2025 – Feb 2025

HONORS & AWARDS

- **VEX U Robotics World Championship**, Finalist (Division Quarter-finalist, Top 8 / 100+ Teams) May 2025
- **China National VEX Robotics Competition**, Second Prize Sep 2024
- **U.S. Mathematical Contest in Modeling (MCM)**, Honorable Mention (H Prize, Top 22%) Jan 2025

PUBLICATION

LATE-Swarm: Late-Bound Semantic-to-Physical Commitment for Natural-Language UAV Swarm Reconfiguration
Yihuan Huang, Hongwei Chen, Jianxiang Wang, JianYu Kong, and Tao Jiang
Submitted to *IEEE International Conference on Robotics and Automation* (ICRA 2027) | **Under review**

RESEARCH EXPERIENCE

LLM-Driven Multi-UAV Swarm Command-to-Control System | Core Developer Mar 2026 – Sep 2026

Advisor: Tao Jiang, Professor, School of Automation, Chongqing University

- Architected LATE-Swarm, a ROS 2/PX4 framework for natural-language UAV swarm control; achieved 100% mission success in 60 Gazebo trials with 8-16 UAVs and 0.085 m mean tracking RMSE.
- Designed a staged pipeline that first interprets formation goals, then determines UAV target positions from live swarm state and sets feasible flight timing from assignment results and motion constraints.
- Developed lexicographic UAV-to-slot optimization using 3D closest-approach analysis, prioritizing hard-conflict avoidance before clearance and travel cost; handled residual execution risks through feedback avoidance.

Visual-Inertial End-Effector Stabilization under Base Motion | Visiting Research Student Jul 2026 – Oct 2026

Advisor: Shiyu Zhao, WINDY Lab, Westlake University

- Developed a ROS 2 perception-control pipeline for 6-DoF end-effector stabilization on a physical robotic arm, integrating visual state estimation and real-time motion compensation to counter external base disturbances.
- Combined D435i stereo-inertial ORB-SLAM3 with joint encoders and calibrated camera-robot extrinsics to reconstruct 6D world poses, accounting for the camera's fixed housing mount and the wrist's independent rotation.
- Designed a 25 Hz Cartesian controller with SO(3) pose errors, damped Jacobian inversion, and velocity feedforward that separates external disturbances from arm self-motion using synchronized vision and encoder feedback.
- Validated compensation under ~10 cm base displacement, achieving 2.3 mm / 0.53° mean final-10-s pose error estimated from SLAM and encoders; sustained compensation during a 217 s demonstration with 3D base motion.

ROS 2-Based Distributed UAV Swarm Formation and Avoidance System | Core Developer Sep 2025 – May 2026

Advisor: Tao Jiang, Professor, School of Automation, Chongqing University

- Designed a distributed multi-UAV formation and avoidance algorithm with an Improved Artificial Potential Field and finite-time state estimator; validated formation tracking and obstacle avoidance in ROS 2/Gazebo simulations.
- Formulated double-tanh saturation to bound formation forces and decoupled leader-follower avoidance, prioritizing collision repulsion to resolve target collapse and deadlocks while preserving stable V-shaped formations.
- Implemented differential-flatness mapping with LADRC to suppress inter-drone downwash and avoid attitude singularities during dynamic maneuvers; achieved steady-state tracking errors below 0.5 m in physical flight tests.

Vision-Based NoMaD Navigation for TITA Wheeled-Legged Robot | Core Developer Oct 2025 – May 2026

Advisor: Can Pu, Associate Professor, National Elite Institute of Engineering, Chongqing University

- Engineered a vision-based topological navigation system for the TITA wheeled-legged robot, enabling autonomous exploration and collision-free trajectory generation in unmapped indoor environments.
- Bridged the Sim2Real domain gap by designing a dynamic observation queue and customizing visual preprocessing pipelines, mapping diffusion-policy outputs directly to real-world physics.
- Developed a high-frequency ROS 2 control node, translating model-predicted spatial waypoints into continuous chassis velocity commands via a customized Pure Pursuit tracking algorithm.
- Validated system robustness through physical deployments on the TITA platform, successfully executing complex dynamic obstacle avoidance without relying on precise metric maps.

PROJECT EXPERIENCE

Vision-Guided Mobile Manipulator for Manipulation & Writing | Course Project Sep 2024 – Jan 2025

Advisor: Long Bai, Professor, School of Mechanical and Vehicle Engineering, Chongqing University

- Developed a vision-guided mobile manipulator with a 4-DOF arm for picking, placing, and character writing; achieved a 95% pick-and-place success rate in a 30 x 30 cm workspace.
- Implemented object detection and line following with OpenMV/Python, enabling the mobile platform to navigate autonomously to target locations and provide visual inputs for manipulation.
- Solved arm inverse kinematics and implemented Arduino/C++ servo control, converting Cartesian targets from the vision system into joint commands for object manipulation and character writing.

LEADERSHIP & TEACHING EXPERIENCE

VEX Robotics Team, Chongqing University | Deputy Lead, Control Systems Group

- Mentored a team of 5 new members in C++ programming and PID control theory for competitive robotics.
- Co-led strategy sessions for competition, analyzing opponents and optimizing autonomous routines.

Department of Robotics Engineering, Chongqing University | Undergraduate Teaching Assistant

- Led hands-on programming sessions for a class of 90+ students, providing direct guidance on STM32 microcontroller development and debugging.

TECHNICAL SKILLS

- **Programming & ML:** C/C++, Python, MATLAB, PyTorch, NLP, OpenCV
- **Robotics & Simulation:** ROS 2, PX4, Gazebo, MATLAB/Simulink, Linux (Ubuntu)
- **Embedded & CAD:** STM32, Arduino, ESP32, FPGA, SPI/UART/I2C; SolidWorks, KiCad, Altium